

Lena Stempfle, PhD

Postdoctoral Researcher



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lena-stempfle

About me

I am a postdoctoral researcher at MIT and the Broad Institute, working on reliable and trustworthy AI systems for healthcare and biomedical data. My research focuses on foundation model evaluation, multimodal and longitudinal health data, clinically grounded AI evaluation, and machine learning under missingness. I have worked on wearable foundation models, memorization and privacy risks in foundation models, interpretable machine learning, and large-scale health prediction systems. I completed my PhD in Computer Science at Chalmers University of Technology and have authored publications at venues including ICML, AISTATS, AAAI, and ML4H, including an ICML 2025 spotlight paper.

Areas of Specialization

Healthcare AI • Foundation Model Evaluation • Multimodal Learning • Machine Learning under Missingness • Interpretable AI • Causal Inference

Machine Learning & AI

PyTorch • TensorFlow • Hugging Face • Scikit-learn • Representation Learning • Statistical Machine Learning

Programming & Infrastructure

Python • R • SQL • Git • Docker • Slurm • HPC • Distributed Computing

EXPERIENCE

- 2025–now **Postdoctoral researcher**
MASSACHUSETTS INSTITUTE OF TECHNOLOGY (MIT) · Cambridge, USA
Research on foundation model evaluation, multimodal learning from longitudinal and wearable health data, and scalable AI methods for biomedical applications. Work includes reliable and trustworthy AI systems, clinically grounded evaluation, and privacy and memorization risks in foundation models.
- 2025–2020 **PhD candidate**
CHALMERS UNIVERSITY OF TECHNOLOGY · Gothenburg, Sweden
Developed interpretable machine learning methods for healthcare data with missing values, published at top-tier conferences, while also contributing to teaching, supervision in collaboration with Sahlgrenska University Hospital, and research community activities at Chalmers University of Technology.
- 2024 **Visiting graduate student**
MASSACHUSETTS INSTITUTE OF TECHNOLOGY (MIT) · Cambridge, USA
Performed and published a meta-survey on human versus large language model performance parity, and researched on tabular LLMs for health outcome prediction using sociodemographic and environmental information.
- 2024 **Visiting graduate student**
PREMEDICAL INRIA-INSERM TEAM · Montpellier, France
Developed a risk score for predictions under missing values in the context of TraumaBase, a network of clinicians from trauma centers across France, and conducted a user study on interpretable machine learning and missing values.
- 2015–2020 **Research Assistant**
KARLSRUHE INSTITUTE OF TECHNOLOGY (KIT) AND FRAUNHOFER INSTITUTE FOR APPLIED INFORMATION TECHNOLOGY (FIT) · Germany
Worked across multiple research projects in digital health, machine learning, human-computer interaction, and information systems, contributing to research, teaching, conference organization, and EU grant writing.
- 2015–2017 **Fulltime Internships in Digital Transformation, Project Management, Controlling, and Audit**
MÜCKE, ROTH & COMPANY, ALLIANZ INSURANCE, KPMG · Germany
Completed internships across consulting, insurance, and audit, focusing on digital healthcare strategy, agile project management, financial planning, and IT portfolio optimization.

References and certificates upon request.

EDUCATION

- 2019 **M.Sc. Information Systems, Law & Management**
Karlsruhe Institute of Technology (KIT), Germany
- 2018 **M.Sc. Computer Science, Exchange Studies**
Chalmers University of Technology
- 2017 **B.Sc. in Business Management**
University of Bayreuth, Germany

TALKS/TEACHING

- 2020–now Reviewer for ML4H, ICES, AIES, ICML conferences.
- 2025 Invited speaker "Interpretable ML and missing values" at Data to Actionable Knowledge (DtAK) group at Harvard, RISE Sweden, KTH and PreMeDiCaL Inria-Inserm team.
- 2025–2020 Lectured and/or assisted for the courses *Introduction to ML*, *Design for AI systems*, *Causality and Causal Inference*, and *Research Methods in Data Science* at Chalmers University of Technology.

AWARDS & GRANTS

- 2022 **Winner of the WASP Secure and Private AI Hackathon**
WASP SUMMER SCHOOL
- 2021, 2024 **Recipient of WASP research exchange grants for research visits to MIT, USA, and Inria-Inserm, France**
WASP / CHALMERS UNIVERSITY OF TECHNOLOGY

Languages

German Mother tongue
English Fluent
Swedish Intermediate

Interests

Climbing • Hiking • Volleyball
• Painting

Publications

*equal contribution, see my
webpage for more
publications

PUBLICATIONS

- 2026 **Evaluation without Generation: Non-Generative Assessment of Harmful Model Specialization with Applications to CSAM**
SURIYAKUMAR, SEKHARI, **LENA STEMPFLE**, WANG, SIMPSON, PORTNOFF, GHASSEMI, WILSON 
• Preprint on Arxiv
- 2026 **Clinically Grounded Privacy Evaluation of Medical LLMs**
RONAGHI, **LENA STEMPFLE**, TONEKABONI, GHASSEMI, ALZENTZER 
• In Submission
- 2026 **Privacy implications of medical AI models**
TONEKABONI*, **STEMPFLE***, RONAGHI, COUPETTE, COHEN, ALSENTZER, GHASSEMI 
• In Submission
- 2026 **Who said it better? The missing human baseline and judge**
STEMPFLE*, HUO*, AHMED, GHASSEMI, SALAUDEEN 
• In Submission
- 2025 **An Investigation of Memorization Risk in Healthcare Foundation Models**
TONEKABONI™, **STEMPFLE***, FALLAHPUR, GERYCH, GHASSEMI 
NeurIPS 2025
- 2025 **Predicting melanoma in the adult Swedish population using ML on registry data**
MARTIN GILLSTEDT, **LENA STEMPFLE**, JOHN PAOLI, FREDRIK D. JOHANSSON, SAM POLESIE 
Acta Dermato-Venereologica, 2026
- 2025 **Prediction models that learn to avoid missing values**
LENA STEMPFLE, ANTON MATSSON, NEWTON MWAI, FREDRIK D. JOHANSSON 
International Conference on Machine Learning (ICML) • Accepted as Spotlight
- 2025 **Applying Machine Learning to High-Dimensional Proteomics Datasets for the Identification of Alzheimer's Disease Biomarkers**
CHRISTOFFER IVARSSON ORRELID, OSCAR ROSBERG, SOPHIA WEINER, JOHAN GOBOM, FREDRIK D. JOHANSSON, NEWTON MWAI, **LENA STEMPFLE** 
Fluids and Barriers of the CNS
- 2024 **MINTY: Rule-based models that minimize the need for imputing features with missing values**
LENA STEMPFLE & FREDRIK D. JOHANSSON 
International Conference on Artificial Intelligence and Statistics. PMLR 964-972)
- 2024 **Learning replacement variables in interpretable rule-based models**
LENA STEMPFLE & FREDRIK D. JOHANSSON 
ICML 3rd Workshop on Interpretable Machine Learning in Healthcare (IMLH)
- 2024 **How should we represent history in interpretable models of clinical policies?**
ANTON MATSSON, **LENA STEMPFLE**, YAOCHEN RAO, ZACHARY R. MARGOLIN, HEATHER J. LITMAN, FREDRIK D. JOHANSSON 
Proceedings of the 4th Machine Learning for Health Symposium, PMLR 259:714-734
- 2024 **Handling missing values in clinical ML: Insights from a large-scale expert study**
LENA STEMPFLE, JAMES ARTHUR, JULIE JOSSE, TOBIAS GAUSS, FREDRIK D. JOHANSSON 
Findings of the 4rd Machine Learning for Health Symposium (ML4H)
- 2023 **Sharing pattern submodels for prediction with missing values**
LENA STEMPFLE, ASHKAN PANANI, FREDRIK D. JOHANSSON 
AAAI Conference on Artificial Intelligence
- 2021 **Predicting progression & cognitive decline in amyloid-positive patients with Alzheimer's disease**
HÁKON VALUR DANSSON, **LENA STEMPFLE**, HILDUR EGILSDÓTTIR, ALEXANDER SCHLIEP, ERIK PORTELIUS, KAJ BLENNOW, HENRIK ZETTERBERG, FREDRIK D. JOHANSSON 
Alzheimer's Research and Therapy 13, 151

VOLUNTEER WORK (SELECTED) & SUMMER SCHOOLS

2026	Organizer of NeurIPS workshop: Privacy risks in foundation models MIT 
2026–now	International Advisory Board Member CENTER FOR NEW BIOMARKERS IN NEURODEGENERATIVE DEMENTIAS 
2021–2025	Organizer of weekly Machine Learning @ DSAI Seminar CHALMERS UNIVERSITY OF TECHNOLOGY 
2024, 2025	Organizer of Workshop for Women in ML CHALMERS UNIVERSITY OF TECHNOLOGY 
2023–2024	Inclusive Leadership Program, Gender Initiative for Excellence (GENIE) CHALMERS UNIVERSITY OF TECHNOLOGY 
2024	Nobel Turing Grand Challenge CHALMERS UNIVERSITY OF TECHNOLOGY 
2021	Summer Schools on Data Science and AI in Health and EEML UNIVERSITY OF GRONINGEN AND BUDAPEST UNIVERSITY OF TECHNOLOGY AND ECONOMICS (BME) 
2019	Founder of <i>she.codes by TEC</i> KIT KARLSRUHE AND TUM MUNICH 